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Measurement and PC-SAFT Modeling of the Solubility of Gallic Acid in Aqueous Mixtures of Deep Eutectic Solvents

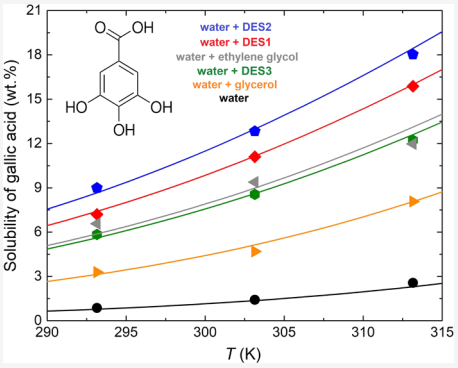
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ABSTRACT: Deep eutectic solvents have appeared as potential solvents for improving the extraction of polyphenols from vegetable or fruit matrixes. Since gallic acid is abundant in these sources, it is considered as a typical standard for quantifying their total polyphenol content after extraction with solvents. However, there are no extensive studies on the solubility behavior of gallic acid in different solvents or deep eutectic solvents. Thus, in this work, the solubility of gallic acid is measured in pure water; aqueous solutions of different hydrogen bond donors such as ethylene glycol, levulinic acid, and glycerol; and aqueous mixtures of deep eutectic solvents using choline chloride as the hydrogen bond acceptor and ethylene glycol, levulinic acid, and glycerol as the hydrogen bond donors. All of the measurements were performed at 293.15, 303.15, and 313.15 K and at 101.3 kPa and were validated by comparing the solubility of gallic acid in water from the literature. Results suggest that a 50 wt % aqueous solution of deep eutectic solvent based on ethylene glycol or glycerol improves the gallic acid solubility compared with a 50 wt % aqueous solution of its corresponding hydrogen bond donor. The deep eutectic solvent containing levulinic acid acts as the best aqueous mixture for gallic acid dissolution. Nondissolved gallic acid was measured after equilibrium using powder X-ray diffraction, showing that its structure does not change upon mixing with all of the liquid mixtures. All of the solid−liquid equilibrium results were accurately modeled with perturbed-chain statistical associating fluid theory (PC-SAFT).



INTRODUCTION

Polyphenols are natural compounds, mainly present in fruits and vegetables, that are recognized as important food or pharmaceutical additives that provide health benefits to humans after their intake. For instance, given their ability to strongly interact with enzymes, they can protect against a number of chronic diseases.[1] Some well-documented health benefits associated with the consumption of phenolic compounds include reducing levels of cholesterol and hypertension and protecting against cardiovascular diseases[2,3] and cancer.[4] Moreover, some polyphenols are antifungal,[5] antimicrobial,[6,7] anti-inflammatory,[8−10] and antioxidant agents.[9,11−13]

Polyphenols are organic compounds that include at least one hydroxyl group attached to one or more benzene rings. They can be classified in flavonoids that are composed of flavonols, flavones, anthocyanidins, etc. and nonflavonoids, such as phenolic alcohols, stilbenes, phenolic acids, etc.[14] These groups are the building blocks of high-molecular-weight polyphenol polymers such as tannins or other macromolecules. Specifically, tannins are classified into hydrolyzable and nonhydrolyzable (condensed), where hydrolyzable tannins are esters of gallic acid (gallo- and ellagitannins) with a central carbohydrate core and condensed tannins (also known as proanthocyanidins) are polymers of polyhydroxyflavan-3-ol monomers structurally related to flavonoids.[1,15−19]

Efficient processes for extracting polyphenols from vegetable or food matrixes are needed to obtain isolated polyphenols or rich polyphenol extracts that can serve as additives for food or pharmaceutical products. Some well-studied and widely applied polyphenol extraction methods, at the laboratory scale or at the industrial level, are conventional solid−liquid extraction with organic solvents or aqueous mixtures at atmospheric pressure, soxhlet, pressurized hot liquid extraction, ultrasound, and microwave among others.[20,21] The selection of an adequate method depends on the properties of the raw material, the intended use, the specific family of polyphenols that are expected to be recovered, etc.

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Table 1. Characteristics of Chemicals Used in This Work

chemical	abbreviation	M (g/mol)	CAS	supplier	type	purity mass fraction[a]
gallic acid	GA	170.12	149-91-7	Sigma-Aldrich		0.975
ethanol	EtOH	46.07	64-17-5	Acros Organics	AcroSeal Extra dry	0.995
choline chloride	ChCl	139.62	67-48-1	Sigma-Aldrich		0.980
ethylene glycol	EG	62.07	107-21-1	VWR Chemicals	AnalaR NORMAPUR	0.997
levulinic acid	Lev	116.12	123-76-2	Sigma-Aldrich		0.980
glycerol	Gly	92.09	56-81-5	Acros Organics	ACS Reagent	0.996

[a]Provided by the supplier.

The typical solvents for conventional solid−liquid extraction of polyphenols are ethanol, glycerol, methanol, ethylene glycol, acetone, etc.[22,23] and their aqueous solutions. Some drawbacks of conventional solvent extraction are low yields, the large volume of solvent needed, is a slow process, and sometimes effective solvents are not environment-friendly. Thus, deep eutectic solvents (DESs) have been tested as an attractive alternative to organic solvents, mainly because of their biodegradability, low cost, and their well-known capacity for extracting phenolic compounds.[22−25] DESs are composed of a hydrogen bond acceptor (HBA), usually choline chloride, and a hydrogen bond donor (HBD), such as glycols, glycerol, organic acids, and other compounds. The idea is to prepare a HBA/HBD combination composed of natural and nonharmful compounds for extracting polyphenols since these extracts are typically used as additives in foods or pharmaceutical products. Thus, it is necessary to avoid traces of any undesirable toxic compound in the polyphenol extract. One of the disadvantages of DESs is their high viscosity compared with conventional organic solvents, for instance, ethanol or acetone.[26] Hence, it is common to observe that water is added to DESs to decrease their viscosity and mass transfer impediments when they are used for extraction processes.[27]

There are a number of studies reporting the extraction of phenolic compounds from vegetable matrixes using not only organic solvents but also DESs and their aqueous mixtures.[25,28,29] Typically, the extract is only analyzed in terms of the total polyphenol content and antioxidant capacity, and, in some cases, the presence of polyphenol standards are tested to quantify them in the liquid phase. A common extraction experimental study comprises different combinations of water + cosolvent mixtures at varying concentrations and temperatures and at a fixed pressure.[27,30] This procedure requires a large number of experiments for optimizing the process, even more, if certain polyphenol standards are required to be present in high concentrations in the aqueous extract. However, some computational studies have recently been performed for selecting solvents that can selectively extract target solutes from a matrix.[31,32] Nonetheless, the thermodynamic analysis of pure polyphenol standards in terms of experimental modeling of their solubility in water or water + cosolvent is scarce, mainly because of the cost and stability of the isolated solute.

Gallic acid is one of the most abundant phenolic compounds in nature. It is commonly present as a building block of hydrolyzable tannins or in free form. It is found as a biologically active component in oil mill wastewater;[33−36] in food grains such as seeds,[37] barley, millet, beans, peas, carobs;[15] in plants, fruits, and vegetables[38] such as olives,[36] apples, bananas, blackberries, blueberries, dates, grapes, hawthorn berries, peaches, pears, persimmons, plums, raspberries, and strawberries;[15] and in liquids such as red wine,[39] tea,[4,9] and olive oil[34,36] among many others. The high availability of gallic acid in these matrixes makes it the common reference to quantify their total polyphenol content in terms of a gallic acid equivalent (mg/L GAE) using the Folin−Ciocalteau assay.[40] Since gallic acid is abundant, has a relatively low cost, and is a stable molecule, there are reports of its solubility in water or aqueous mixtures with some organic solvents.[41−52] From our knowledge, there are no reports of gallic acid solubility in aqueous DES mixtures along with the potential changes in the crystalline structure after mixing with the liquid phase. However, there has been a recent study reporting the solubility of gallic acid and other polyphenols in a DES based on choline chloride + xylitol.[53]

The modeling of the aqueous solubility of gallic acid has been typically performed with empirical methods,[42,45,46,51] but there is a report using the cubic-plus-association equation of state.[43] Thus, state-of-the-art equations of state or activity coefficient models are required to accurately describe the solubility of polyphenolic solutes in a liquid phase. A promising method for modeling solid−liquid equilibrium of natural compounds in a solvent or a mixture of them is perturbed-chain statistical associating fluid theory (PC-SAFT).[54,55] This equation of state is capable of modeling complex systems, for instance, the solubilities of vitamins in water[56] and lactose in aqueous electrolyte mixtures,[57] the solid−liquid behavior of deep eutectic solvents,[58] the solubility of peptides in aqueous solutions of ionic liquids,[59] etc.

This work aims to measure the solubility of gallic acid in water and aqueous mixtures of different HBDs, such as ethylene glycol, levulinic acid, and glycerol, and three DESs formed by choline chloride as the HBA and the same HBDs mentioned before, all in a molar ratio of 1:2. The gallic acid structure is assessed before and after the measurements through powder X-ray diffraction (PXRD) to verify its stability under the presence of a liquid phase. All of the solid−liquid equilibrium measurements are modeled by PC-SAFT to obtain pure-component parameters for gallic acid that could be transferable for modeling its behavior when solubilized in complex mixtures.

EXPERIMENTAL SECTION

Materials. Gallic acid, choline chloride, and levulinic acid were purchased from Sigma-Aldrich; ethanol and glycerol from Acros Organics; and ethylene glycol from VWR Chemicals. All chemicals were used as received. Milli-Q water (Milli-Q Reference Merck, conductivity approximately <100 μS/cm) was used to prepare all of the solvent mixtures. A summary of all of the compounds used in this work along with their characteristics is shown in Table 1.

Preparation of DESs. All of the DESs were prepared gravimetrically using a Mettler Toledo analytical balance (Excellence Model XS205DU, 0.0001 g uncertainty). Initially,

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HBA and HBD were placed in a flask that was tightly sealed and then heated to 353.15 K. The sample was stirred until a homogeneous liquid phase was formed. The amount of water in each DES was measured using a Karl Fischer Coulometer (831 KF, Metrohm, Switzerland). The HBA was choline chloride, and the three HBDs used were ethylene glycol (DES1), levulinic acid (DES2), and glycerol (DES3). All of the samples were prepared in a HBA/HBD mole ratio of 1:2, and their water content was below 0.1 wt %. Finally, water was added to the DES in a flask placed on the analytical balance until an aqueous solution composed of 50 wt % of DES was obtained. The solvent mixture composed of water + HBD was mixed gravimetrically until the aqueous sample had a 50 wt % of HBD.

Solubility Measurements. Solid solubilities in the aqueous solutions were measured using the saturated shake-flask method. The procedure consists of the addition of the pure solvent or a mixture of solvents to a 2 mL Eppendorf tube containing enough gallic acid to avoid its complete dissolution in the liquid phase. Three tubes with the same solid + liquid mixture were prepared and placed in a thermoshaker (Eppendorf ThermoMixer C, ±0.1 °C) and stirred at 1000 rpm for 48 h, followed by a settling time of another 48 h. All this was performed at a controlled constant temperature set in the thermoshaker. After the equilibrium, a sample of 10 μL was taken with a micropipette from the liquid phase of each flask and diluted for sampling. Then, the absorbance of the sample was measured using an ultraviolet−visible (UV−vis) spectrophotometer (Analytik JenaSpecord 210 Plus) at a wavelength of 213 nm, and the amount of gallic acid dissolved was quantified by comparing the absorbance measurement of the diluted sample with a calibration curve. This procedure was performed in quadruplicate for each system. The pH value was measured for all of the saturated liquid phases before solute quantification.

Powder X-ray Diffraction Measurements. The solid residue of gallic acid that remains inside the Eppendorf tube after the equilibrium measurements was analyzed through PXRD using a MiniFlex X-ray diffractometer (Rigaku, Tokyo, Japan) and compared with pure gallic acid powder. Measurements were carried out at 298 K and 101.3 kPa, scanning between 2 and 60° at a speed of 5°/min and with a step size of 0.02°. Cu Kα radiation was applied (K =1.54184 nm c) with a tube voltage and current set at 40 kV and 15 mA, respectively, in an analogous procedure, as shown elsewhere.[60,61]

THEORY

PC-SAFT was proposed by Gross and Sadowski[54,55] as a modification of SAFT using hard chains as the reference system. PC-SAFT is based on the perturbation theory as different contributions to the residual Helmholtz energy (a^{res}), as shown in eq 1

$$a^{res} = a^{hc} + a^{disp} + a^{assoc} \tag{1}$$

where a^{hc} represents the Helmholtz energy related to the hard chain contribution, a^{disp} is related to the dispersive attraction of nonspherical molecule contribution, and a^{assoc} is related to the association contributions. For nonassociative molecules, only three parameters of pure components are required: the segment diameter (σ_i), the number of segments per chain (m_i), and the dispersion energy parameter (u_i/k). Two additional parameters are needed for associative molecules: the associating energy $\epsilon^{A_iB_i}$ and the effective associating volume $\kappa^{A_iB_i}$. All of these are typically fit to density and vapor pressure data.

Cross-interactions between two molecules i and j are calculated using the conventional Berthelot−Lorenz combination rules, as shown in eqs 2 and 3

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j) \tag{2}$$

$$u_{ij} = \sqrt{u_i u_j}(1 - k_{ij}) \tag{3}$$

where k_{ij} is the adjustable binary interaction parameter fit to experimental data when necessary. In a mixture containing two associating compounds, the cross-associating interactions are quantified with the expressions proposed by Wolbach and Sandler[62] in eqs 4 and 5

$$\epsilon^{A_iB_j} = \frac{\epsilon^{A_iB_i} + \epsilon^{A_jB_j}}{2} \tag{4}$$

$$\kappa^{A_iB_j} = \sqrt{\kappa^{A_iB_i}\kappa^{A_jB_j}}\left(\frac{\sqrt{\sigma_{ii}\sigma_{jj}}}{\frac{1}{2}(\sigma_{ii} + \sigma_{jj})}\right)^3 \tag{5}$$

The solubility of gallic acid in the liquid phase can be quantified through the solid−liquid equilibrium relationship shown in eq 6, assuming a pure solid phase and neglecting the influence of the difference of heat capacities of the pure solid and the liquid, as follows

$$x_i^L = \frac{\varphi_{0i}^L}{\varphi_i^L} \exp\left\{-\frac{\Delta h_{0i}^{SL}}{RT}\left(1 - \frac{T}{T_{0i}^{SL}}\right)\right\} \tag{6}$$

where $\varphi_{0\,i}^L$ is the fugacity coefficient of the pure component and φ_i^L is the fugacity coefficient of component i (polyphenol) in the mixture. Also, Δh_{0i}^{SL} and T_{0i}^{SL} are the enthalpy and temperature of melting of the pure polyphenol, respectively, and T is the temperature of the system. Melting properties for gallic acid, i.e., melting temperature and melting enthalpy, were taken from Mota et al.[43] who report that these values are equal to 499 K and 38 770 J/mol, respectively. They also report that these properties were estimated with the Marrero and Gani group contribution method,[63] since gallic acid decomposed upon melting and no reliable experimental values can be obtained. The required fugacity coefficients are calculated using PC-SAFT as shown in eq 7

$$RT \ln \varphi_i = \left(\frac{\partial n a^{res}}{\partial n_i}\right)_{T,V,n_{k\neq i}} - RT \ln Z \tag{7}$$

where n_i and n are the moles of component i and the total moles in the mixture, respectively, R is the universal gas constant, and Z is the compressibility factor. Moreover, Z is also calculated with PC-SAFT, as shown in eq 8

$$Z = 1 + \rho\left(\frac{\partial(a^{res}/RT)}{\partial\rho}\right)_{T,n_i} \tag{8}$$

where ρ is the density at a given pressure and temperature, as calculated elsewhere.[64]

RESULTS AND DISCUSSION

Solubility of Gallic Acid in Water. The solubility of gallic acid was measured in pure water at 293.15, 303.15, and 313.15

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K and at 101.3 kPa. These results were compared with literature values,[42,43,45−52] as shown in Figure 1a. Also, the

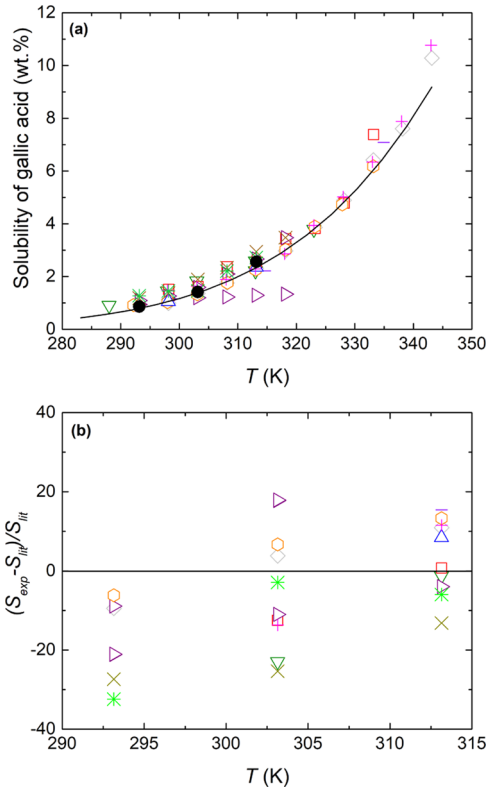


Figure 1. (a) Solubility of gallic acid in water (wt %) at different temperatures and at 101.3 kPa and (b) percentage deviations $((S_{exp} - S_{lit})/S_{lit})$ of experimental data from this work (S_{exp}) with the literature (S_{lit}). This work, solid black circle; Noubigh et al.,[41,44,47] open right-pointing triangle; Lu and Lu,[42] open diamond; Mota et al.,[43] open inverted triangle; Daneshfar et al.,[45] open square; Srinivas et al.,[46] dash; Li et al.,[48] open hexagon; Vilas-Boas et al.,[49] open triangle; Dabir et al.,[50] plus; Dali et al.,[51] cross; and Aydi et al.,[52] asterisk. The continuous line represents the PC-SAFT model with parameters reported in Table 5.

percentage deviation $(100 \times (S_{exp} - S_{lit})/S_{lit})$ for gallic acid solubility was determined for some literature measurements (S_{lit}) compared with our experimental results (S_{exp}) at the same temperatures as observed in Figure 1b. Literature values that are in agreement with our results are reported by Lu and Lu[42] with deviations below 10.94%, by Noubigh et al.[47] below 10.98%, and Li et al.[48] below 13.32%, all of these for the three temperatures reported in our work. The largest deviation is observed with values reported by Noubigh et al.[41,44] at 313.15 K and was not considered in Figure 1b because it was beyond the error reported by the other studies. The solubility of gallic acid in pure water increases with increasing temperature. The pH of the saturated water solution is shown in Table 2, and the experimental points for the solubility of gallic acid in pure water are shown in Table 3.

The solubility of gallic acid was also measured in pure ethanol, for comparison, at the same temperature and pressure conditions as in water. These results were not considered for validation or comparison with the other solubility measurements because of the results obtained in the PXRD, as explained below.

Table 2. pH Measurements and pH Standard Deviation (SD) of the Liquid Phase after Equilibrium with Gallic Acid

solvent	pH	SD
H_2O	2.91	0.006
ethylene glycol[a]	2.57	0.042
glycerol[a]	2.76	0.020
DES1[a]	2.27	0.038
DES2[a]	1.78	0.055
DES3[a]	2.13	0.041

[a]50 wt % in water.

Table 3. Experimental Solubility (S_{exp}), Standard Deviation of Experimental Solubility (S_{SD}), and Solubility Calculated with PC-SAFT ($S_{PC\text{-}SAFT}$) of Gallic Acid in the Liquid Phases at Temperatures between 293.15 and 313.15 K and at 101.3 kPa[a]

T (K)	solubility (wt % of gallic acid)			
	S_{exp}	S_{SD}	$S_{PC\text{-}SAFT}$	deviation[b]
	Water			
293.15	0.861	0.009	0.776	−9.914
303.15	1.413	0.033	1.361	−3.688
313.15	2.558	0.097	2.301	−10.038
	50 wt % Ethylene Glycol + 50 wt % Water			
293.15	6.567	0.111	5.873	−10.573
303.15	10.325	0.266	8.986	−4.294
313.15	11.974	0.164	13.113	9.506
	50 wt % Glycerol + 50 wt % Water			
293.15	3.287	0.130	3.132	−4.708
303.15	4.691	0.202	5.130	9.342
313.15	8.075	0.515	8.062	−0.161
	50 wt % DES1 + 50 wt % Water			
293.15	7.198	0.336	7.241	5.653
303.15	11.106	0.385	10.862	0.404
313.15	15.875	0.194	15.494	0.748
	50 wt % DES2 + 50 wt % Water			
293.15	8.997	0.052	8.668	−3.655
303.15	12.833	0.437	12.979	1.138
313.15	18.031	0.260	18.442	2.282
	50 wt % DES3 + 50 wt % Water			
293.15	5.839	0.123	5.611	−3.914
303.15	8.564	0.272	8.614	0.584
313.15	12.224	0.231	12.606	3.125

[a]Standard uncertainties u are $u(T) = 0.1$ K and $u(P) = 1$ kPa. The relative standard uncertainty u_r is $u_r(w_1) = 0.05$, where w_1 represents the wt % of gallic acid in the liquid solution. Combined expanded uncertainty U $(w_2) = 0.2$ with a 0.95 level of confidence, where w_2 represents the wt % of DES or HBD in water. [b]Deviation $= 100 \times \left(\frac{S_{PC-SAFT} - S_{exp}}{S_{exp}}\right)$

Solubility of Gallic Acid in Aqueous Solutions of HBD or DES. The solubility of gallic acid was also measured in 50 wt % aqueous mixtures of ethylene glycol, levulinic acid, glycerol, DES1, DES2, and DES3 at temperatures of 293.15, 303.15, and 313.15 K and at 101.3 kPa. All of the pH values of the saturated liquid mixtures are reported in Table 2, and the experimental solubility values are shown in Table 3. Ethylene glycol, levulinic acid, and glycerol are typical HBDs for preparing DESs used for extracting polyphenols.[25] Also, DESs are usually mixed with water when used as extracting solvents close to ambient temperature to decrease their viscosity.[27] Thus, the solubility study of gallic acid was performed at the

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same concentration of water for assessing the effect of using only the aqueous HBD or the aqueous DES on gallic acid dissolution in the liquid mixture. However, the UV−vis spectrum of gallic acid was distorted when an aqueous solution of levulinic acid was mixed.[65] This is probably due to a strong interaction or reaction of levulinic acid with gallic acid because of the long contact time (4 days) of both components. For this reason, only ethylene glycol and glycerol were used as HBD precursors for gallic acid solubility studies. However, DES2 prepared with levulinic acid as the HBD did not affect the UV−vis spectrum of gallic acid when this compound was solubilized in this aqueous phase.

Figure 2 shows the gallic acid solubility comparison in the aqueous solutions of ethylene glycol and glycerol, observing

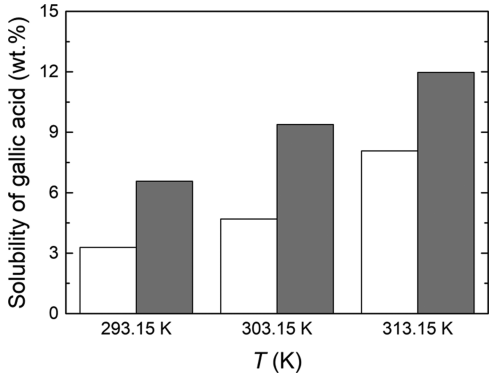


Figure 2. Solubility of gallic acid in aqueous solutions of 50 wt % ethylene glycol (gray bar) and glycerol (white bar) at 293.15, 303.15, and 313.15 K and at 101.3 kPa.

that at the three temperatures, the gallic acid concentration increases in the liquid phase with increasing temperature and when using ethylene glycol instead of glycerol. At all of the temperatures, the concentration of gallic acid is 3.3−5.6 wt % higher when using ethylene glycol in the aqueous phase compared with glycerol.

On the other hand, Figure 3 shows the solubility of gallic acid in the aqueous mixtures of DES1, DES2, and DES3. Again, the solubility of gallic acid increases at higher temperatures, and using DES, the following order is observed: DES3 < DES1 < DES2. Thus, DES2, with levulinic acid as the HBD, provides the highest concentration of gallic acid in the liquid phase with a maximum solubility of 18.03 wt %,

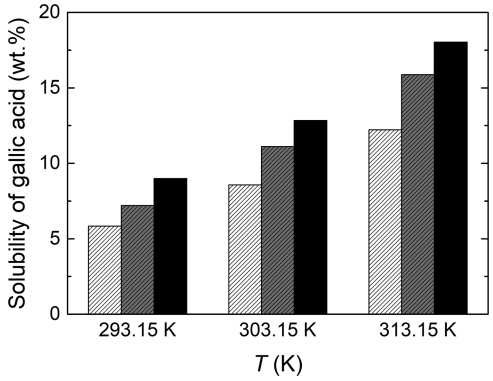


Figure 3. Solubility of gallic acid in aqueous solutions of 50 wt % of DES1 (gray bar with patterns), DES2 (black bar), and DES3 (white bar with patterns) at 293.15, 303.15, and 313.15 K and at 101.3 kPa.

compared with 15.88% in DES1 and 12.22 wt % in DES3 at 313.15 K.

Figure 4 shows the comparison of gallic acid solubility in the aqueous mixtures of DES and their respective aqueous HBDs

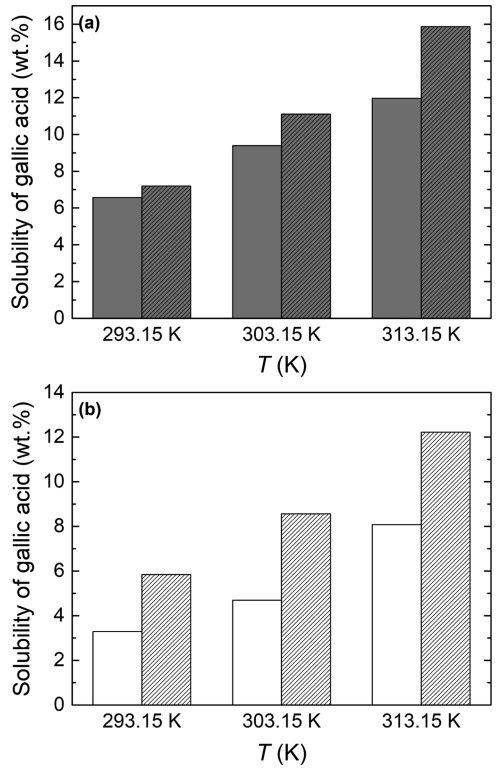


Figure 4. Solubility of gallic acid in (a) ethylene glycol[a] (gray bar) and DES1[a] (gray bar with patterns) and (b) glycerol[a] (white bar) and DES3[a] (white bar with patterns) at 293.15, 303.15, and 313.15 K and at 101.3 kPa, where ([a]) represents aqueous solutions in 50% weight fraction.

at the same water content. Figure 4a shows the comparison of DES1 with ethylene glycol and Figure 4b, DES3 with glycerol. In general, the aqueous DES mixture yields a higher solubility compared with the aqueous HBD solution at all of the temperatures. Then, the addition of choline chloride to the HBD improves the ability to dissolve gallic acid in the aqueous solution. The analysis of the experimentally observed solubility dependencies on each solvent is discussed in the next section.

Solvent Influence on Gallic Acid Solubility. The experimental results are contrasted with the Kamlet−Taft solvatochromic parameters of the different solvents used in this work to assess the behavior of the solubility curves. Kamlet−Taft parameters provide a quantitative analysis of the properties of solvents, for instance, for tuning them for interacting with a specific solute[66] or to understand their hydrogen-bonding or polarizability characteristics that are closely related with the dissolution of a target compound.[67]

The three Kamlet−Taft parameters for a specific solvent are hydrogen bond donor capacity (α), hydrogen bond acceptor capacity (β), and the polarity/polarizability (π^*) properties of solvents as contributors to the overall polarity of the solvent. Kamlet−Taft parameters for the solvent used in this work were obtained from the literature[66−70] and are shown in Table 4. They were correlated in Figure 5a in terms of β vs π^* and in Figure 5b as β vs α. As observed in Figure 5a, the solubility

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Table 4. Solvatochromic Parameters for Molecular Solvents

solvent	α	β	π^*
water	1.17[66]	0.14[66]	1.09[66]
ethanol	0.86[66]	0.75[66]	0.84[66]
ethylene glycol	0.90[68]	0.52[68]	0.92[68]
levulinic acid	0.55[69]	0.45[69]	0.58[69]
glycerol	0.90[66]	0.51[70]	1.07[66]
DES1	1.47[67]	0.57[67]	1.07[67]
DES2	0.51[67]	0.57[67]	1.00[67]
DES3	1.49[67]	0.52[67]	1.11[67]

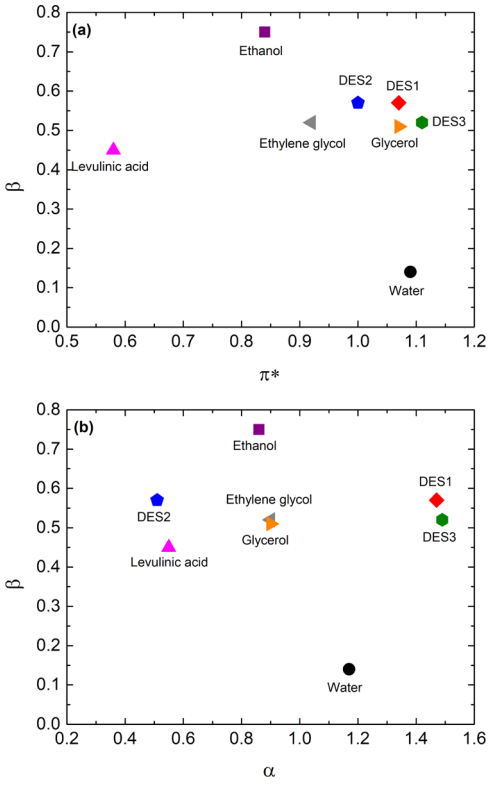


Figure 5. Solvatochromic parameters (a) β vs π^* and (b) β vs α for water (circle), ethanol (square), ethylene glycol (left-pointing triangle), levulinic acid (triangle), glycerol (right-pointing triangle), DES1 (diamond), DES2 (pentagon), and DES3 (hexagon).

values of gallic acid in the aqueous mixtures follow a similar trend as the values of β of the pure solvents. The higher solubility of gallic acid in ethylene glycol compared with glycerol, both with a similar value of α and β, is shown as a difference in the polarity of both molecules, i.e., the lower π^* of ethylene glycol leads to an increase in the solubility. Also, the internal interactions of the OH-groups of glycerol produce a decrease in the interactions with the solute, leading to a decrease in its solubility compared with ethylene glycol, as explained elsewhere.[71] Gallic acid shows a maximum pH of 2.91 in the aqueous phase, as reported in Table 2. Then, all of the pH values are lower than the pK_a of the most acidic functional group in gallic acid (pK_a = 4.25 for the carboxyl group[72]). Therefore, the formation of the species in equilibrium is favored in gallic acid, therefore acting as a HBD. Also, a density functional theory (DFT) study showed that the carboxylic acid of gallic acid (that acts as a HBD) interacts with the OH^- of the solvents (that act as a HBA) when ethylene glycol, glycerol, and other alcohols are used.[73]

Indeed, there are reports where gallic acid is used as a HBD for the preparation of DESs with choline chloride.[74] Since gallic acid has a predominantly HBD behavior, the best solvents for dissolving gallic acid must act as the best hydrogen bond acceptors. Therefore, the best acceptors present a higher value of β and better affinity with gallic acid. Figure 6 shows the

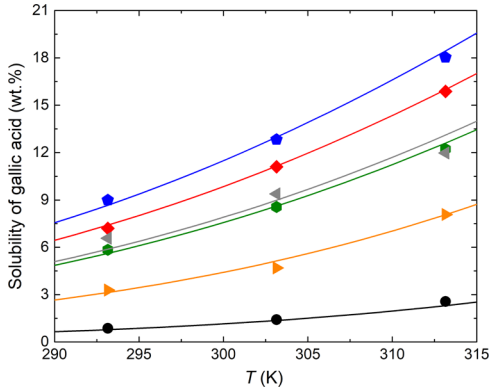


Figure 6. Solubility of gallic acid in water (circle), ethylene glycol[a] (left-pointing triangle), glycerol[a] (right-pointing triangle), DES1[a] (diamond), DES2[a] (pentagon), and DES3[a] (hexagon) at 293.15, 303.15, and 313.15 K and at 101.3 kPa. Continuous lines represent the PC-SAFT model with parameters reported in Table 5. ([a]) represents aqueous mixtures at 50 wt% of the solvent.

comparison of solubility of gallic acid in all of the solvents used in this work, showing that the gallic acid concentration increases in the aqueous phase in the same order as the values of β for the different solvents, as reported in Figure 5a.

Parameter π^* apparently has a significant effect on the solubility at similar values of α and β, and the correlation between the solvents and the experimental solubility with a variation of π^* can be observed by comparing Figures 5a and 6, which has been previously explained in the ethylene glycol and glycerol solubility comparison. This can be observed in Figure 5, where DES1 and DES2 have the same value of β but very different values of α. Here, the solubility of gallic acid in DES2 is higher compared with DES1, and π^* is lower for DES2 compared with DES1. Finally, the solubility of gallic acid does not show a correlation with parameter α, since gallic acid behaves like a HBD and the solubility data should be related to the opposite parameter, which is to accept hydrogen bonds.

PXRD Analysis. PXRD analysis of the nondissolved fraction of solid gallic acid was performed after the solid−liquid equilibrium measurements to assess changes in polyphenol after its contact with the liquid phase. Figure 7 shows the diffractogram of pure gallic acid compared with those of the solid gallic acid remaining after its mixture with water; ethanol; and all of the aqueous solvent solutions composed of ethylene glycol, glycerol, DES1, DES2, and DES3. These results show that gallic acid has main peaks at 16, 19, 25, and 26° 2θ, which are comparable to those reported by Kaur et al.[75] It is observed that the PXRD peaks for gallic acid remain unchanged after mixing in almost all of the liquid phases, presenting the same crystalline structure as pure polyphenol without mixing. The only exception is observed when the solute is mixed with pure ethanol, where its diffraction curve shows a decrease and a slight displacement in the main peak by 16° 2θ. Also, two new peaks appear at approximately 25 and 28° 2θ. This could be explained by the

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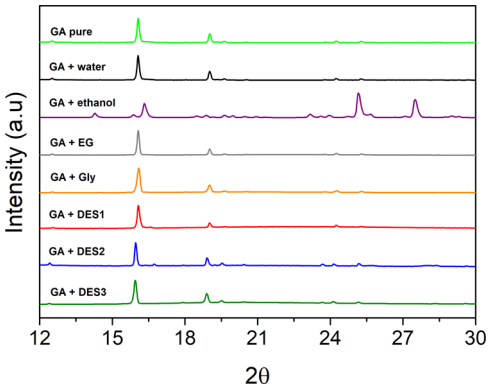


Figure 7. PXRD of pure gallic acid (light green line) and gallic acid after equilibrium in water (black line); ethanol (purple line); and aqueous solutions of ethylene glycol (gray line), glycerol (orange line), DES1 (red line), DES2 (blue line), and DES3 (green line).

strong interaction that occurs between gallic acid and pure ethanol that could form a polymorph, a solvate, or others. Although we measured the solubility of gallic acid in ethanol, these results cannot be compared with other systems because of the change in the crystal structure of gallic acid. Therefore, they were not included in this work and should not be further analyzed.

PC-SAFT Modeling of the Solubility of Gallic Acid. The solubility of gallic acid in all of the pure liquids and aqueous mixtures was modeled using PC-SAFT by calculating the fugacity coefficients from eq 6. Pure-component parameters for gallic acid were fit to its experimental solubility in water obtained in this work. Gallic acid was modeled as a self-associating molecule by assuming three association donor and acceptor sites. Pure-component parameters for the other compounds used in this study (water, ethylene glycol, levulinic acid, glycerol, DES1, DES2, and DES3) were retrieved from the literature.[55,76−81] All of the PC-SAFT pure-component parameters are reported in Table 5. Most of the binary interaction parameters (k_{ij}) were fit in this work from solid−liquid equilibrium and some values, for instance, water + choline chloride or water + levulinic acid, were taken from the literature.[79,80] All of the k_{ij} values are reported in Table 6, where they are represented by eq 9 with the constants a_{ij} and b_{ij} as

$$k_{ij} = a_{ij} + b_{ij} \times (T - 298.15\ \mathrm{K}) \quad (9)$$

Table 6. PC-SAFT Binary Interaction Parameters k_{ij} Used in This Work

system	a_{ij}^{*}	b_{ij}^{**}
GA/H_2O[a]	−0.0095	
GA/ChCl[a]	−0.2500	
GA/EG[a]	−0.2100	
GA/Lev[a]	−0.2900	
GA/Gly[a]	−0.1600	
GA/DES1[a]	−0.175	
GA/DES2[a]	−0.2120	
GA/DES3[a]	−0.2000	
H_2O/ChCl[80]	−0.0285	0.0001
H_2O/EG[a]	−0.0300	
H_2O/Lev[79]	0.0600	
H_2O/Gly[a]	−0.0050	

[a]Type of data used for fitting, solid−liquid (this work).

Only the binary systems composed of water + choline chloride needed a temperature-dependent k_{ij}.

The PC-SAFT modeling was performed with two different approaches: (a) modeling the DES as a pseudopure compound and (b) modeling with the individual components of the DESs. Since some interaction parameters between the individual components themselves were not available in the literature, the missing k_{ij} values were fitted in this work. The k_{ij} values were not changed through the different DES systems. Both approaches show reliable results, which are in good agreement with the experimental data. For reasons of presentation only, the results for the PC-SAFT modeling with the pseudo-pure-component parameters are shown in this work.

The PC-SAFT modeling results are compared with the experimental values of the solubility of gallic acid in the different liquid phases, as shown in Figure 6. The deviations ($100 \times (S_{\text{PC-SAFT}} - S_{\text{exp}})/S_{\text{exp}}$) of the solubility calculated with PC-SAFT ($S_{\text{PC-SAFT}}$) compared with all of the experimental points (S_{exp}) are reported in Table 3, obtaining values within 10.6% and showing that PC-SAFT presents a reasonable representation of the experimental data within the experimental error observed for data from this work compared with the literature.

CONCLUSIONS

Solubility measurements for gallic acid were carried out at 293.15, 303.15, and 313.15 K and at 101.13 kPa in seven different liquid phases: in pure water for comparing with the literature; in binary mixtures composed of 50 wt % of different

Table 5. PC-SAFT Parameters

PC-SAFT parameters used in this work						
abbreviation	m_i	σ_i [Å]	ϵ/k [K]	$\kappa^{A_iB_i}$	$\epsilon^{A_iB_i}/\kappa$ [K]	scheme
GA[a]	3.1897	2.2488	185.9531	0.0110	1722.8110	(3-3)
H_2O[76]	1.2047	2.7927	353.9449	0.0451	2425.6710	2B (1-1)
ChCl[80]	13.0200	2.3680	228.0700	0.2000	8000.0000	2B (1-1)
EG[78]	3.3940	2.8052	319.9200	0.0228	1889.8600	4C (2-2)
Lev[78]	2.0311	4.1241	266.4953	0.0171	4578.3660	2B (1-1)
Gly[81]	2.0070	3.8150	430.8200	0.0019	4633.4700	2B (1-1)
DES1[77]	3.5408	3.2782	465.0741	0.0021	4971.9890	2B (1-1)
DES2[77]	3.4672	3.6847	477.7234	0.0030	4601.3090	2B (1-1)
DES3[82]	7.7594	2.5699	275.0000	0.1000	5000.0000	2B (1-1)

[a]Fit to experimental data from this work.

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HBDs such as ethylene glycol, levulinic acid, and glycerol with water; and 50 wt % of DES mixtures composed of choline chloride and the previously mentioned HBDs at a 1:2 mole ratio in water. Solubility data in water was in agreement with those of three previous studies[42,47,48] with deviations below 13.5% at all of the temperatures. Also, our experimental data shows the best agreement with the literature at 313.15 K with deviations below 14% with nine different studies.[42,43,45,47−52] All of the experimental values show that the solubility of gallic acid increases in the liquid phase on increasing the temperature and follows the following order: aqueous DES2 > aqueous DES1 > aqueous ethylene glycol ≈ aqueous DES3 > aqueous glycerol > pure water.

Solubility results were analyzed in terms of the Kamlet−Taft solvatochromic parameters of the solute and the different solvents used for the experiments. Since gallic acid has pK_a values lower than the pH of the medium, the formation of species in equilibrium is favored by gallic acid acting as a hydrogen bond donor. Then, the best solvents for solubilizing gallic acid have to act as the best hydrogen bond acceptors (or higher β parameter). Thus, the solubility sequence in the different solvents mainly corresponds to the order observed for the β and π^* parameters. On the other hand, parameter α does not have a strong influence on the solubility of gallic acid in the different solvents.

Pure and nondissolved gallic acid were analyzed through PXRD after the equilibrium experiments to observe the influence of the solvent on the diffraction planes of the solute. Pure gallic acid presented the same diffraction curve for all of the systems indicating that there was no change in the structure of the unit cell of the polyphenol even under the presence of very dissimilar liquid phases. However, the diffraction curve changed in the presence of ethanol. The solubility results of gallic acid in ethanol were not used in this work for comparison with other systems.

PC-SAFT modeling was performed using the solid−liquid equilibrium equation, obtaining a good representation of gallic acid in the different solvents with deviations within 10.04% for pure water, 10.57% in aqueous HBD mixtures, and 5.65% in aqueous DES mixtures.

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Notes

The authors declare no competing financial interest.

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